

5-Bar SCARA v2 Onboarding Package

M4

Encoder + Sensor Fusion Edition | Engineering Roadmap

COMMS & INTEGRATION LEAD

1 | ROLE & OBJECTIVE

As **Comms & Integration Lead (M4)**, you are the connective tissue of the system. You own two complementary phases: **Phase 4** — design, implement, and validate the extended **ESP-NOW FusionPacket protocol** that carries Kalman filter output (position + velocity + confidence) from the vision node to the kinematic node; and **Phase 5** — unite all six team members' subsystems into a single, calibrated, operational closed-loop pick-and-place cycle.

Primary Mission: Deliver a <20-byte FusionPacket with CRC-16 integrity, <2% packet loss, and end-to-end latency under 15ms; then orchestrate the 9-point pixel-to-world calibration and verify the full system runs the pick-and-place cycle with encoder correction and Kalman fusion active simultaneously.

2 | HARDWARE & COMPONENTS REQUIRED

Component	Specification / Notes	Qty
ESP32-CAM (Vision Node)	M3's firmware flashed; acts as ESP-NOW transmitter	1
ESP32 Dev Board (Kinematic Node)	M2's firmware flashed; acts as ESP-NOW receiver	1
WiFi Spectrum Analyzer (optional)	Check for 2.4GHz channel congestion near workspace	1
Calibration Dot Grid (3×3)	Printed at known world coordinates; for 9-point calibration	1
USB-to-Serial Adapter	For flashing and Serial monitoring of both nodes simultaneously	2
Multimeter / Oscilloscope	Power rail verification during integration; relay timing check	1
Laptop with dual Serial monitors	Monitor both ESP32 nodes simultaneously during integration	1

3 | SOFTWARE & PLATFORMS REQUIRED

Tool / Library	Purpose	Source
Arduino IDE 2.x / PlatformIO	Develop and flash Phase 4 comms firmware	arduino.cc

Tool / Library	Purpose	Source
ESP-NOW (built-in Espressif)	Peer-to-peer 2.4GHz wireless; <1ms latency; no router needed	ESP32 core
CRC-16 implementation (Arduino)	Validate packet integrity; reject corrupted bytes	Library or manual
Python + pyserial + pandas	Log 1000 packets for loss rate measurement (TEST 4-C)	pip install
Serial Monitor (both nodes)	Watch g_target fields update in real time; monitor confidence gating	Arduino IDE
Git	Coordinate FusionPacket header changes with M2 and M3	git-scm.com
Spreadsheet	9-point calibration: record measured vs. true coordinates, compute RMS	Excel / Sheets

4 | AI INTEGRATION STRATEGY

Your work requires protocol design precision, CRC implementation, and system-level integration debugging. Use AI to validate packet structure, generate test harnesses, and diagnose integration failures:

Prompt 1 — CRC-16 implementation review:

"I need a CRC-16 (polynomial 0x8005 or CCITT 0x1021) implementation in Arduino C++ for a 20-byte FusionPacket struct. The CRC covers bytes 0-17 (first 18 bytes). Provide the crc16() function, show how to compute it at transmit side, verify at receive side, and explain why __packed__ struct is required. Also show the static_assert to verify sizeof(FusionPacket)==20 at compile time."

Prompt 2 — Debug packet loss during integration:

"My ESP-NOW link shows ~5% packet loss in testing (1000 packet test, 50 dropped). Both nodes are 1.5 meters apart. I'm in a lab with other WiFi devices on 2.4GHz. What are the top 5 causes of ESP-NOW packet loss and how do I diagnose each? Should I change ESP-NOW channel? How does the sequence number dedup affect perceived loss rate?"

Prompt 3 — 9-point calibration RMS calculation:

"I have 9 calibration points with known world coordinates (Xw,Yw) and measured ESP-NOW packet values (Xm,Ym). The errors are: [list your actual measurements]. Calculate RMS error. If RMS > 3mm, determine whether the error pattern suggests camera tilt (rotation error), scale drift (uniform scaling), or nonlinear distortion. Provide the corrective formula for each case."

Prompt 4 — Integration sequence plan:

"I am integrating a 5-bar SCARA robot with these subsystems: ESP32-CAM running Kalman filter, ESP32 kinematic node with AccelStepper and encoder ISR, ESP-NOW FusionPacket link, and an electromagnet relay. List the exact integration sequence (which firmware to flash first, which Serial logs to verify at each step) to minimize debugging time. What are the top 3 failure modes at each integration step?"

5 | BASIC ARCHITECTURE & CONNECTIONS

Connects From	Your Role	Delivers To
M3: Kalman output (X,Y,Vx,Vy,confidence)	Pack into FusionPacket with CRC-16 and sequence byte	M2: ESP-NOW receiver ISR

Connects From	Your Role	Delivers To
M2: confidence threshold gate	Document threshold values (>180 move, <80 idle) in firmware	Phase 5 integration test
M3: pixel-to-world calibration	Run 9-point grid; verify RMS < 3mm; update scale params	M3: update firmware constants
M2: encoder correction output	Verify correction fires post-move; log encoder error pre/post	M5: TEST 5-C sign-off
M5: electromagnet timing	Log relay energize/de-energize encoder positions	M5: interference check
All members: Phase 5 integration	Flash, connect, calibrate, and run single pick cycle end-to-end	M6: final validation handoff

Packet Flow: M3 transmit_fused() → [ESP-NOW 2.4GHz] → on_fusion_rcv() ISR (M4 implements on kinematic node) → CRC check → dedup check → g_target struct → M2 state machine

6 | TIMELINE & MILESTONES

[DAY 1]

- Sync FusionPacket struct definition with M2 and M3 — agree on all fields and byte order
- Verify ESP-NOW peer pairing works: both nodes exchange a test packet (ping/pong)
- Implement CRC-16 function; verify with a known test vector
- Review M3's Phase 3 output (SortPacket→FusionPacket migration path)

[DAYS 2–3 (Phase 4 — FusionPacket Protocol)]

- Implement full FusionPacket struct with #pragma pack and static_assert
- Implement on_fusion_rcv() ISR with CRC guard and sequence dedup
- Implement confidence gating: >180 start move, <80 stay IDLE
- TEST 4-A: static_assert(sizeof(FusionPacket)==20) passes on BOTH projects
- TEST 4-B: End-to-end latency < 5ms average, < 15ms worst case (1000 packet log)
- TEST 4-C: Packet loss < 2% over 1000 packets
- TEST 4-D: Inject corrupted byte — verify CRC rejection, packet never accepted
- TEST 4-E: Confidence field correctly gates IDLE/MOVING transitions
- M3 reviews Phase 4 — obtain sign-off

[DAYS 4–6 (Phase 5 — Full System Integration)]

- Flash M3's Phase 3B firmware to ESP32-CAM; confirm Kalman output on Serial
- Flash M2's Phase 2B firmware to ESP32; confirm encoder counts at home
- Verify ESP-NOW link active; confirm g_target.confidence field populated on kinematic node
- Run single manual pick by Serial-injecting coordinates; verify encoder correction triggers
- Connect live ESP-NOW feed; verify confidence gating suppresses spurious picks
- Verify Vx,Vy non-zero in FusionPacket when object placed with a push
- Run 9-point calibration with M2 and M3; compute RMS; update scale parameters if needed
- TEST 5-A: Static pick accuracy — 4/5 trials electromagnet contacts object
- TEST 5-B: 20-cycle repeatability — ≥18/20 success at same position
- TEST 5-C: Encoder correction measurably improves accuracy (log both states)
- TEST 5-D: Fused position smoother than raw centroid over 50 frames (log both)
- TEST 5-E: Out-of-workspace target rejected, IDLE preserved

7 | ESCALATION MATRIX

Issue / Blocker	First Contact	Second Contact
ESP-NOW pairing fails between nodes	Verify both on same WiFi channel; hardcode peer MAC addresses	M6 — hardware/environment check
Packet loss > 2% in open space	Check for 2.4GHz channel congestion; use AI Prompt 2	M5 — retest in controlled environment
FusionPacket CRC mismatch at receiver	Verify #pragma pack on BOTH sides; check endianness	M2 — sync struct definition
Confidence gating causes phantom IDLEs	Lower confidence threshold from 80; check M3 occlusion_frames logic	M3 — recalibrate confidence formula
9-point calibration RMS > 3mm	Check camera tilt; use AI Prompt 3 for error pattern diagnosis	M3 — update scale_x, scale_y
Encoder correction doesn't trigger in Phase 5	Verify POSITION_ERROR_THRESHOLD; check correction bypassed by speed	M2 — review correction layer
Integration Phase 5 gate blocked	Coordinate with M2 and M3 jointly; M6 for arbitration	M6 for escalation

You are the architect who brings six independent systems together into one cohesive machine. When Phase 5 runs its first end-to-end pick-and-place cycle, that's your masterpiece. Keep everyone aligned, keep the packets flowing, and trust the process. ■